

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12401204
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Boggs; Matthew Lee et al.

Circuit to drive electrically controlled switches within electrified machine

Abstract

A driving circuit to control/drive a contactor using a high-side driver integrated into the battery pack controller. The driver circuit may electrically isolate the controller pins of the battery pack controller (which interface the contactors) from the low-voltage digital processing core of the battery pack controller. The driver circuit may provide techniques to diagnose a fault by monitoring the voltage at the output of the driver as compared to the reference voltage. The driver circuit may include an electronic switch, such as a field-effect transistor, that is configured on the high side of the contactor to control voltage from an energy source to the contactor.

Inventors: Boggs; Matthew Lee (Metamora, IL), Miller; Jason Lee (Princeville, IL)

Applicant: Caterpillar Inc. (Peoria, IL)

Family ID: 1000008779982

Assignee: Caterpillar Inc. (Peoria, IL)

Appl. No.: 17/855968

Filed: July 01, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240006893 A1	Jan. 04, 2024

Publication Classification

Int. Cl.: H02J7/00 (20060101); B60R16/033 (20060101)

U.S. Cl.:

CPC **H02J7/0013** (20130101); **B60R16/033** (20130101); **H02J7/00304** (20200101);
 H02J7/0047 (20130101);

Field of Classification Search

CPC: H02J (7/0013)

USPC: 320/128

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7605568	12/2008	Gangstoe et al.	N/A	N/A
9583745	12/2016	White et al.	N/A	N/A
10155449	12/2017	Chen et al.	N/A	N/A
2013/0207464	12/2012	Kluthe	307/10.7	H01M 10/613
2015/0340743	12/2014	Fink	429/61	G01R 31/396
2016/0141894	12/2015	Beaston	N/A	N/A
2017/0166075	12/2016	Hong et al.	N/A	N/A
2018/0233928	12/2017	Li	N/A	H02J 7/0029
2019/0103750	12/2018	Kristensen	N/A	N/A
2020/0412155	12/2019	Choi	N/A	H02J 7/007182
2021/0141043	12/2020	Bertness	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
105490255	12/2016	CN	N/A
209462020	12/2018	CN	N/A
209766327	12/2018	CN	N/A
112216558	12/2020	CN	N/A
112216558	12/2020	CN	N/A

OTHER PUBLICATIONS

Written Opinion and International Search Report for Int'l. Patent Appln. No. PCT/US2023/068658, mailed Oct. 10, 2023 (36 pgs). cited by applicant

Primary Examiner: Fantu; Yalkew

Background/Summary

TECHNICAL FIELD

(1) This document relates generally to a battery management system for an electrified machine and, in particular, to techniques for operating electrically controlled switches.

BACKGROUND

(2) Battery Management Systems (BMS) for electrified machines may operate electrically-controlled high-current and high-voltage switches called contactors for the purpose of

connecting/disconnecting strings of lithium-ion batteries and for controlling the charging/discharging of the batteries within a battery pack on a mobile machine or stationary power application.

(3) CN112216558A describes a relay driving circuit and an electrical system, the relay driving circuit comprises a power supply module, a control module, a high-side driving module and a low-side driving module. The high-side driving module and the low-side driving module are connected in series and are respectively controlled by the control module, so that a high-side and low-side series driving mode of the relay is realized. The problem of misoperation of the relay when the output end of the existing single-side relay driving circuit driven by a single high-side switch or a single low-side switch is short-circuited or open-circuited can be avoided.

SUMMARY OF THE INVENTION

(4) This disclosure describes a driving circuit to control/drive a contactor using a high-side driver integrated into the battery pack controller. The driver circuit may electrically isolate the controller pins of the battery pack controller (which interface the contactors) from the low-voltage digital processing core of the battery pack controller. The driver circuit may provide techniques to diagnose a fault by monitoring the voltage at the output of the driver as compared to the reference voltage. The driver circuit may include an electronic switch, such as a field-effect transistor, that is configured on the high side of the contactor to control voltage from an energy source to the contactor.

(5) In some aspects, this disclosure is directed to a battery management system configured to couple to a plurality of battery modules, wherein corresponding ones of the plurality of battery modules include a battery cell and a cell monitoring circuit, the battery management system comprising: a plurality of electrically controlled switches electrically coupled to the plurality of battery modules; a switch control circuit configured to operate the plurality of electrically controlled switches; driver circuitry electrically coupled to the switch control circuit and configured to drive the switch control circuit, the driver circuitry electrically coupled to a first voltage source; and processing circuitry electrically coupled to a second voltage source that is different from the first voltage source, wherein the processing circuitry is configured to receive battery information from corresponding cell monitoring circuits of the plurality of battery modules, wherein the processing circuitry is electrically isolated from the driver circuitry, and wherein the processing circuitry is in communication with the driver circuitry via at least one data channel.

(6) In some aspects, this disclosure is directed to a method of operating a battery management system having a plurality of battery modules, wherein corresponding ones of the plurality of battery modules include a battery cell and a cell monitoring circuit, the method comprising: coupling driver circuitry to a first voltage, wherein the driver circuitry is electrically coupled to and configured to drive a switch control circuit, and wherein a switch control unit is configured to operate a plurality of switches electrically coupled to the plurality of battery modules; coupling processing circuitry to a second voltage that is different from the first voltage; electrically isolating the processing circuitry from the driver circuitry, wherein the processing circuitry is in communication with the driver circuitry via at least one data channel; and generating, using the processing circuitry, a first driver control signal to enable an electronic switch of the driver circuitry via the at least one data channel.

(7) In some aspects, this disclosure is directed to a electrified machine comprising: a machine frame; and a battery management system supported by the machine frame and configured to couple to a plurality of battery modules, wherein corresponding ones of the plurality of battery modules include a battery cell and a cell monitoring circuit, the battery management system comprising: a plurality of electrically controlled switches electrically coupled to the plurality of battery modules; a switch control circuit configured to operate the plurality of electrically controlled switches; driver circuitry electrically coupled to the switch control circuit and configured to drive the switch control circuit, the driver circuitry electrically coupled to a first voltage source; and processing circuitry

electrically coupled to a second voltage source that is different from the first voltage source, wherein the processing circuitry is configured to receive battery information from corresponding cell monitoring circuits of the plurality of battery modules, wherein the processing circuitry is electrically isolated from the driver circuitry, and wherein the processing circuitry is in communication with the driver circuitry via at least one data channel.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of an example of an electrified machine that may implement various techniques of this disclosure.
- (2) FIG. 2 is an example of a battery system that may implement various techniques of this disclosure.
- (3) FIG. 3 is a block diagram of an example of the driver circuitry of FIG. 2 that may implement various techniques of this disclosure.
- (4) FIG. 4 is a block diagram of another example of the driver circuitry of FIG. 2 that may implement various techniques of this disclosure.
- (5) FIG. 5 is a flow diagram of an example of a method of operating a battery management system having a plurality of battery modules using various techniques of this disclosure.

DETAILED DESCRIPTION

- (6) Examples according to this disclosure are directed to techniques for driving electrically controlled switches within an electrified machine.
- (7) Electrified machines are powered by batteries instead of an engine. A battery pack may include one or more battery modules, and a battery module may include one or more battery cells. Battery modules may include a battery cell monitoring circuit to monitor, among other things, a voltage and a temperature of the battery cells in the battery module.
- (8) A battery pack controller may be in communication with the various battery module monitoring circuits to receive and aggregate the information from the various battery modules. The battery pack controller may coordinate operation of all the battery modules. The battery modules may be connected, such as in series and in parallel, with one another using high voltage, high current electrically controlled contacts of a contactor.
- (9) Some battery management approaches drive, or control, these contactors using pilot/interposed relays, which are external to both the contactor and the battery pack controller or with low-side, sinking driver circuits within the battery pack controller. The present inventors have recognized that these approaches may have various drawbacks, such as increased component count, increased cost, and/or low reliability. In addition, these approaches may also cause incompatibility with the operation of some contactor devices. The present inventors have recognized a need for a solution that drives/controls the contactors using a high-side driver, not a low-side driver, that is integrated into the battery pack controller and that does not require external pilot/interpose relays.
- (10) This disclosure describes a driving circuit to control/drive a contactor using a high-side driver integrated into the battery pack controller. The driver circuit may electrically isolate the controller pins of the battery pack controller (which interface the contactors) from the low-voltage digital processing core of the battery pack controller. The driver circuit may provide techniques to diagnose a fault by monitoring the voltage at the output of the driver as compared to the reference voltage. The driver circuit may include an electronic switch, such as a field-effect transistor, that is configured on the high side of the contactor to control voltage from an energy source to the contactor.
- (11) The driver circuit may include a current sensing circuit between the electronic switch and the energy source to monitor the current through the electronic switch and provide a sensing current

output to a protection circuit. The protection circuit may turn the electronic switch off when the sensing current exceeds either a high-current threshold or a low-current threshold for a specific period of time. Further, in the event of a fast turn-off of the driver circuit after receiving the off command, the driver circuit design may provide sufficient development of back EMF for the contactor,

(12) FIG. 1 is a perspective view of an example of an electrified machine that may implement various techniques of this disclosure. A non-limiting example of an electrified machine **100** is shown in FIG. 1. The electrified machine **100** may be any machine or vehicle powered, at least partially, by batteries, as will be described below with reference to FIG. 2. The electrified machine **100** may be a mining truck, as depicted, or any other on-highway or off-highway machine or vehicle. In some examples, the electrified machine **100** may include a machine frame **102** supporting at least an operator control station **104** and a dump body **106**. Ground-engaging elements **108**, such as wheels and/or endless-link treads, may also be supported by the machine frame **102**, and may be powered by a drive system, discussed below, supported by the machine frame **102**.

(13) FIG. 2 is an example of a battery system that may implement various techniques of this disclosure. The battery management system **200** may include a battery module **202** that includes a plurality of battery cells **204A-204N** (collectively referred to in this disclosure as “battery cells **204**”). The battery cells **204** may interface with a cell monitoring circuit **206** that may monitor various battery parameters, such as voltage and temperature of the battery cells. The cell monitoring circuit **206** may manage the health of the battery cells **204**, including charging and discharging of the battery cells **204**. The battery module **202** may be coupled in series, for example, with other battery modules **202** to create a “DC Link”, such as 700 volts (V) or greater, which is represented by DC Link+ **208A** and DC Link-**208B**.

(14) The DC Link+ **208A** may be coupled via an RC network **210A** to a machine frame **212**, such as the machine frame **102** of FIG. 1, by a DC bonding mutual point. For example, the RC network **210A** may AC couple the floating differential DC Link to a pseudo ground reference. The DC Link-**208B** may be coupled via a similar RC network **210B** to the machine frame **212**.

(15) The battery management system **200** may include a battery pack controller **214** to aggregate and control the activity of the various battery modules **202** on the electrified machine, such as the electrified machine **100** of FIG. 1, which may include a hundred or more battery cells **204**. The battery pack controller **214** may communicate with the various cell monitoring circuits **206** via datalink **216** (for brevity, only one such monitoring circuit **206** has been shown).

(16) The battery pack controller **214** may include processing circuitry **218** and driver circuitry **220**, such as in the same controller housing. The processing circuitry **218** may be electrically isolated from the driver circuitry **220** and in communication with the driver circuitry **220** via one or more data channels **222**. Isolating the processing circuitry **218** from the driver circuitry **220** such that the only communication between them is by data channel **222** protects the processing circuitry **218** from excessively high voltage.

(17) In some examples, the processing circuitry **218** may include one or more datalinks **224**, e.g., Control Area Network (CAN) datalink, a digital processing core **226**, low voltage input/output (I/O) **228**, and one or more sensor supplies **230** to provide power to various sensors external to the battery pack controller **214**. The processing circuitry **218** may be coupled to a first voltage source, such as from a first voltage source **232**, e.g., a 9-32V lead acid battery system.

(18) The driver circuitry **220** may be coupled to a second voltage source, such as a second voltage source **236**, where the second voltage source is different from the first voltage source. Such a design allows the driver circuitry **220** to be powered by an independent, regulated voltage supply on the electrified machine, e.g., the electrified machine **100** of FIG. 1.

(19) The driver circuitry **220** may include contactor control circuitry **234** and be electrically coupled to a switch control circuit **238** of a contactor **240**. A contactor is similar to a relay, with a

coil and switches, e.g., contacts, that operate when the coil is energized or deenergized.

(20) The switch control circuit **238** includes a control coil and feedback circuit and is configured to operate a plurality of switches **242**, e.g., high current, high voltage contacts, that are electrically coupled to a string of battery modules **244**. The string of battery modules **244** includes a plurality of battery modules **202** coupled in series with one another. In some examples, a first contact of the contactor may be coupled near the top of the string of battery modules **244** and a second contact of the contactor may be coupled near the bottom of the string of battery modules **244**. This allows switching of particular strings of battery modules in and out of the DC Link, selectively isolating a string if there is a fault, and/or selectively charging strings.

(21) The battery pack controller **214** outputs a control signal **246** that controls operation of the plurality of switches **242**, e.g., contacts, coupled to the battery modules **244** by driving a control coil of the switch control circuit **238** of the contactor **240** using the control signal **246**. The contactor **240** provides a feedback signal **248** to the battery pack controller **214** to communicate what the contactor **240** believes to be the state (open or closed) of the switches **242**, e.g., contacts. The feedback signal **248** may be an analog signal or a digital signal.

(22) The battery management system **200** may be implemented in an electrified machine, such as the electrified machine **100** of FIG. 1.

(23) FIG. 3 is a block diagram of an example of the driver circuitry **220** of FIG. 2 that may implement various techniques of this disclosure. Like in FIG. 2, the processing circuitry **218** is coupled to the first voltage source **232** and the driver circuitry **220** is coupled to the second voltage source **236**. The processing circuitry **218** is electrically isolated from the driver circuitry **220** using isolation circuitry **300**, e.g., a digital isolator circuit. The processing circuitry **218** and the driver circuitry **220** may communicate over the isolation barrier over one or more data channels **222**.

(24) The driver circuitry **220** may include a high side electronic switch **302**, such as a semiconductor switch, e.g., field-effect transistor (FET), positioned between the second voltage source **236** and the control coil input of the contactor. The control coil input of the external contactor is connected to a product connector via a wiring harness. The electronic switch **302** controls the application of voltage to the contactor from the second voltage source **236**. In some examples, the electronic switch **302** is a p-type FET.

(25) In some examples, the electronic switch **302** has an avalanche rating that allows the contactor control coil to develop the back EMF needed for fast turn-off when the processing circuitry **218** commands the driver circuitry **220** to turn off. Back EMF is needed for the contactor to quickly open its contact. If the avalanche rating of the electronic switch **302** is insufficient, the contactor may be unable to quickly open its contact and break the high voltage, high current connection, which may result in arcing and possibly welding the contacts together.

(26) The driver circuitry **220** may include current sensing circuitry **304** to monitor a current through the electronic switch **302**. In particular, the current sensing circuitry **304** is configured to sense a representation of the current through the electronic switch **302**, such as a voltage generated across a sense resistor.

(27) The driver circuitry **220** in FIG. 3 may include a voltage monitoring circuit **306** configured to monitor a voltage at an output of the driver circuitry **220**, such as at the product connector **308**, and communicate a representation of the voltage to the processing circuitry **218** across the isolation barrier via data channel(s) **222**. For example, the voltage monitoring circuit **306** may allow a comparison of a representation of a voltage on the pins of an interface to the control coils of the contactor.

(28) In some examples, the voltage monitoring circuit **306** includes a comparator circuit coupled to a fixed reference voltage, such as generated by a voltage divider circuit. The voltage monitoring circuit **306** compares an output of the driver circuitry **220** to the fixed reference voltage and provides a corresponding signal to the processing circuitry **218** indicating whether voltage is above or below the fixed threshold voltage. In such an implementation, the voltage monitoring circuit **306**

may communicate a state and not the voltage itself.

(29) The driver circuitry **220** may include protection circuitry **310**. The protection circuitry **310** outputs a control signal to turn OFF, e.g., disable, the electronic switch **302** if the sensed current through the electronic switch **302** either: a) exceeds a high-current threshold or b) exceeds a lower-current threshold for a specific duration of time. In other words, the protection circuitry **310** may disable the electronic switch **302** when a level of current exceeds a first threshold or disable the electronic switch **302** when the level of current exceeds a second threshold for a duration of time, where the first threshold is higher than the second threshold. If either of these protection criteria have been met, the protection circuitry **310** outputs another control signal to deactivate high-side switch control circuitry **312**, which overrides a first driver control signal **316** from the contactor control circuitry **234** of FIG. 2, and latches the electronic switch **302** in an OFF state using latching circuitry **314**.

(30) In some examples, during normal operation, the processing circuitry **218** outputs a first driver control signal **316** to enable the electronic switch **302** via control signal **322**. The first driver control signal **316** turns ON a first n-type FET of the high-side switch control circuitry **312** that pulls down a gate terminal of the electronic switch **302**, e.g., a p-type FET, using control signal **322** to keep the electronic switch **302** turned ON. But, if the protection circuitry **310** detects a fault, the protection circuitry **310** applies a signal to a gate terminal of a second n-type FET of the latching circuitry **314**. When ON, the second n-type FET of the latching circuitry pulls down the gate terminal of the first n-type FET using a control signal **318**, which, in turn prevents the processing circuitry **218** from turning ON the first n-type FET, thereby preventing the electronic switch **302** from turning ON. In this manner, the latching circuitry **314** latches the electronic switch **302** in an OFF state when the level of the current exceeds a threshold.

(31) In some examples, the latching circuitry **314** includes a latch RC circuit timer. When the processing circuitry **218** outputs a second driver control signal **320** to turn OFF the driver circuitry **220**, the latch RC circuit timer begins to discharge until the latching circuitry **314** is reset. In this manner, the second driver control signal **320** disables the electronic switch **302** of the driver circuitry and resets the latching circuitry **314**.

(32) In some examples, two signals may be sent across the isolation barrier via the one or more data channels **222** to the processing circuitry **218**: 1) an ON/OFF signal that controls whether the high side driver circuit is ON/OFF; and 2) a fault feedback signal from the voltage monitoring circuit **306**. Based on this combination, the processing circuitry **218** may determine whether certain faults have occurred. If the processing circuitry **218** determines that a fault occurred, the processing circuitry **218** may transmit, using the one or more data channels **222**, a single ON/OFF control signal to reset the protection latch in the high-side switch control circuitry **312**.

(33) FIG. 4 is a block diagram of another example of the driver circuitry **220** of FIG. 2 that may implement various techniques of this disclosure. Many of the components in FIG. 4 are similar to those shown and described above with respect to FIG. 3 and, for purposes of conciseness, will not be described in detail again. In FIG. 4, the driver circuitry **220** includes fault determination circuitry **400**. When a fault occurs, the protection circuitry **310** outputs a signal **402** to the fault determination circuitry **400**, which determines the fault type, e.g., overcurrent fault and an undercurrent fault. The fault determination circuitry **400** then generates a fault feedback signal **404** representing the fault type to the processing circuitry **218** through the isolation circuitry **300**. In this manner, the fault determination circuitry **400** provides an independent fault feedback signal to the processing circuitry **218** from the driver circuitry **220**. This is in contrast with the techniques of FIG. 3 where voltage comparison information is provided by the voltage monitoring circuit **306** to the processing circuitry **218**, which knows whether it commanded the driver circuitry **220** on/off, for diagnosing the different fault types.

(34) FIG. 5 is a flow diagram of an example of a method **500** of operating a battery management system having a plurality of battery modules using various techniques of this disclosure.

Corresponding ones of the plurality of battery modules include a battery cell and a cell monitoring circuit.

(35) At block **502**, the method **500** includes coupling driver circuitry to a first voltage, wherein the driver circuitry is electrically coupled to and configured to drive a switch control circuit, and wherein a switch control unit is configured to operate a plurality of switches electrically coupled to the plurality of battery modules.

(36) At block **504**, the method **500** includes coupling processing circuitry to a second voltage that is different from the first voltage.

(37) At block **506**, the method **500** includes electrically isolating the processing circuitry from the driver circuitry, wherein the processing circuitry is in communication with the driver circuitry via at least one data channel.

(38) At block **508**, the method **500** includes generating, using the processing circuitry, a first driver control signal to enable an electronic switch of the driver circuitry via the at least one data channel.

(39) The method **500** may optionally include monitoring a voltage at an output of the driver circuitry and communicating the voltage to the processing circuitry via the at least one data channel.

(40) The method **500** may optionally include outputting a first driver control signal to enable an electronic switch of the driver circuitry.

(41) The method **500** may optionally include sensing a representation of a current through the electronic switch.

(42) The method **500** may optionally include receiving the representation of the current through the electronic switch and disabling the electronic switch when a level of the current exceeds a threshold.

(43) The method **500** may optionally include disabling the electronic switch when the level of the current exceeds a first threshold or disabling the electronic switch when the level of the current exceeds a second threshold for a duration of time, wherein the first threshold is higher than the second threshold.

(44) The method **500** may optionally include latching the electronic switch in an OFF state when the level of the current exceeds the threshold.

(45) The method **500** may optionally include outputting a second driver control signal to disable the electronic switch of the driver circuitry and to reset latching circuitry.

INDUSTRIAL APPLICABILITY

(46) Electrified machines are powered by batteries instead of an engine. A battery pack may include one or more battery modules, and a battery module may include one or more battery cells. Battery modules may include a battery cell monitoring circuit to monitor, among other things, a voltage and a temperature of the battery cells in the battery module.

(47) A battery pack controller may be in communication with the various battery module monitoring circuits to receive and aggregate the information from the various battery modules. The battery pack controller may coordinate operation of all the battery modules. The battery modules may be connected, such as in series and in parallel, with one another using high voltage, high current electrically controlled contacts of a contactor.

(48) Some battery management approaches drive, or control, these contactors using pilot/interposed relays, which are external to both the contactor and the battery pack controller or with low-side, sinking driver circuits within the battery pack controller. The present inventors have recognized that these approaches may have various drawbacks, such as increased component count, increased cost, and/or low reliability. In addition, these approaches may also cause incompatibility with the operation of some contactor devices. The present inventors have recognized a need for a solution that drives/controls the contactors using a high-side driver, not a low-side driver, that is integrated into the battery pack controller and that does not require external pilot/interpose relays.

(49) This disclosure describes a driving circuit to control/drive a contactor using a high-side driver

integrated into the battery pack controller. The driver circuit may electrically isolate the controller pins of the battery pack controller (which interface the contactors) from the low-voltage digital processing core of the battery pack controller. The driver circuit may provide techniques to diagnose a fault by monitoring the voltage at the output of the driver as compared to the reference voltage. The driver circuit may include an electronic switch, such as a field-effect transistor, that is configured on the high side of the contactor to control voltage from an energy source to the contactor.

(50) Unless explicitly excluded, the use of the singular to describe a component, structure, or operation does not exclude the use of plural such components, structures, or operations or their equivalents. The use of the terms “a” and “an” and “the” and “at least one” or the term “one or more,” and similar referents in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The use of the term “at least one” followed by a list of one or more items (for example, “at least one of A and B” or one or more of A and B”) is to be construed to mean one item selected from the listed items (A or B) or any combination of two or more of the listed items (A and B; A, and B; A, B and B), unless otherwise indicated herein or clearly contradicted by context. Similarly, as used herein, the word “or” refers to any possible permutation of a set of items. For example, the phrase “A, B, or C” refers to at least one of A, B, C, or any combination thereof, such as any of: A; B; C; A and B; A and C; B and C; A, B, and C; or multiple of any item such as A and A; B, B, and C; A, A, B, C, and C; etc.

(51) The above detailed description is intended to be illustrative, and not restrictive. The scope of the disclosure should, therefore, be determined with references to the appended claims, along with the full scope of equivalents to which such claims are entitled.

Claims

1. A battery management system configured to couple to a plurality of battery modules, wherein corresponding ones of the plurality of battery modules include a battery cell and a cell monitoring circuit, the battery management system comprising: a plurality of electrically controlled switches electrically coupled to the plurality of battery modules; a switch control circuit configured to operate the plurality of electrically controlled switches; driver circuitry electrically coupled to the switch control circuit and configured to drive the switch control circuit, the driver circuitry electrically coupled to a first voltage source; and processing circuitry electrically coupled to a second voltage source that is different from the first voltage source, wherein the processing circuitry is configured to receive battery information from corresponding cell monitoring circuits of the plurality of battery modules, wherein the processing circuitry is electrically isolated from the driver circuitry, and wherein the processing circuitry is in communication with the driver circuitry via at least one data channel.
2. The battery management system of claim 1, further comprising: a voltage monitoring circuit configured to monitor a voltage at an output of the driver circuitry and communicate a representation of the voltage to the processing circuitry via the at least one data channel.
3. The battery management system of claim 1, wherein the processing circuitry is configured to enable the driver circuitry via the at least one data channel.
4. The battery management system of claim 3, wherein the processing circuitry configured to enable the driver circuitry via the at least one data channel is configured to: output a first driver control signal to enable an electronic switch of the driver circuitry.
5. The battery management system of claim 4, further comprising: current sensing circuitry to sense a representation of a current through the electronic switch.
6. The battery management system of claim 5, further comprising: protection circuitry configured to receive the representation of the current through the electronic switch and disable the electronic

switch when a level of the current exceeds a threshold.

7. The battery management system of claim 6, wherein the protection circuitry configured to disable the electronic switch when a level of the current exceeds a threshold is configured to: disable the electronic switch when the level of the current exceeds a first threshold or disable the electronic switch when the level of the current exceeds a second threshold for a duration of time, wherein the first threshold is higher than the second threshold.

8. The battery management system of claim 6, further comprising: latching circuitry to latch the electronic switch in an OFF state when the level of the current exceeds the threshold.

9. The battery management system of claim 8, wherein the processing circuitry is configured to: output a second driver control signal to disable the electronic switch of the driver circuitry and to reset the latching circuitry.

10. The battery management system of claim 1, further comprising: fault determination circuitry configured to generate a fault feedback signal representing a fault type.

11. A method of operating a battery management system having a plurality of battery modules, wherein corresponding ones of the plurality of battery modules include a battery cell and a cell monitoring circuit, the method comprising: coupling driver circuitry to a first voltage, wherein the driver circuitry is electrically coupled to and configured to drive a switch control circuit, and Wherein a switch control unit is configured to operate a plurality of switches electrically coupled to the plurality of battery modules; coupling processing circuitry to a second voltage that is different from the first voltage; electrically isolating the processing circuitry from the driver circuitry, wherein the processing circuitry is in communication with the driver circuitry via at least one data channel; and generating, using the processing circuitry, a first driver control signal to enable an electronic switch of the driver circuitry via the at least one data channel.

12. The method of claim 11, further comprising: monitoring a voltage at an output of the driver circuitry and communicating the voltage to the processing circuitry via the at least one data channel.

13. The method of claim 12, further comprising: outputting a first driver control signal to enable an electronic switch of the driver circuitry.

14. The method of claim 13, further comprising: sensing a representation of a current through the electronic switch.

15. The method of claim 14, further comprising: receiving the representation of the current through the electronic switch and disabling the electronic switch when a level of the current exceeds a threshold.

16. The method of claim 15, further comprising: disabling the electronic switch when the level of the current exceeds a first threshold or disabling the electronic switch when the level of the current exceeds a second threshold for a duration of time, wherein the first threshold is higher than the second threshold.

17. The method of claim 15, further comprising: latching the electronic switch in an OFF state when the level of the current exceeds the threshold.

18. The method of claim 17, further comprising: outputting a second driver control signal to disable the electronic switch of the driver circuitry and to reset latching circuitry.

19. An electrified machine comprising: a machine frame; and a battery management system supported by the machine frame and configured to couple to a plurality of battery modules, wherein corresponding ones of the plurality of battery modules include a battery cell and a cell monitoring circuit, the battery management system comprising: a plurality of electrically controlled switches electrically coupled to the plurality of battery modules; a switch control circuit configured to operate the plurality of electrically controlled switches; driver circuitry electrically coupled to the switch control circuit and configured to drive the switch control circuit, the driver circuitry electrically coupled to a first voltage source; and processing circuitry electrically coupled to a second voltage source that is different from the first voltage source, wherein the processing

circuitry is configured to receive battery information from corresponding cell monitoring circuits of the plurality of battery modules, wherein the processing circuitry is electrically isolated from the driver circuitry, and wherein the processing circuitry is in communication with the driver circuitry via at least one data channel.

20. The electrified machine of claim 19, wherein the battery management, further comprises: a voltage monitoring circuit configured to monitor a voltage at an output of the driver circuitry and communicate a representation of the voltage to the processing circuitry via the at least one data channel.
